Apple Developer Academy Senac

Challenge 6

Grupo 5

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Engage

Engage

Big Idea

Apple Technologies and Frameworks

Essential Question

How to use CoreNFC and AppClips.

Challenge

Develop a Proof of Concept (PoC) to demonstrate the use of CoreNFC and App Clips technologies.

Engage

Challenge statement

Development of a Proof of Concept (PoC) including the implementation of Core NFC and App Clips technologies, to facilitate everyday actions that involve bringing the phone close to trigger a quick task.

Exploratory cycle - technologies apple and frameworks

Exploratory cycle - technologies apple and frameworks

Guiding Questions

What Apple frameworks are available?

What is the purpose of each Apple framework, and what can we do with it?

What Apple technologies are available?

What is the purpose of each Apple technology, and what can we do with it?

Exploratory cycle - technologies apple and frameworks

The Process

- In-depth research of articles.
- Discussion of ideas among team members.
- Validation of the proposal with the mentor.

The Decision

- Core NFC: Chosen as the main Framework
- AppClips: Selected as the ideal Apple Technology
- They could work well together!

Exploratory cycles Investigate

Exploratory cycles Investigate



PoC



CoreNFC



App Clip

Key Findings

Key Findings

PoC

A PoC is a practical experiment used to validate the technical feasibility of an idea before full development. Its purpose is not to create the final product, but to test hypotheses, identify potential difficulties, and reduce technical risks. In our Challenge, we decided to test Core NFC and App Clips separately before integrating them, ensuring that their main functionalities were technically feasible.

Key Findings

CoreNFC

Core NFC is an Apple framework that allows applications to read and write information on compatible NFC tags. Its implementation involves enabling the NFC capability in Xcode, configuring the *info.plist*, and creating a reading session using *NFCNDEFReaderSession*. It supports information such as text, URLs, and other NDEF records, requiring compatible devices and tags, as well as proper data and error handling.

Key Findings

App Clip

App Clips allow users to access specific features of an application without having to install its full version. They are implemented through a separate target in Xcode, associated with a main application. They can be launched through different methods, such as NFC tags, QR Codes, App Clip Codes, and links, using invocation URLs and App Store Connect configurations to direct users to the desired experience.

Synthesis

Synthesis

The Process

- Technical research and discussions.
- Mentor and team validation.
- PoC development and documentation.

The Decision

- App Clips for instant access.
- NFC as a complementary interaction.
- Senac Library as the initial application.

Challenge response

Challenge response

Development of an experience flow for the Senac service ecosystem, initially focusing on the Library and using Apple's native technologies: App Clips and Core NFC. The proposal enables students to access book rentals instantly through an NFC-enabled kiosk and rent books by scanning the NFC tags attached to their covers, reducing user effort and eliminating queues. The scalable experience can also be applied to the Gym and Sports Center, with the Library defined as the MVP.

ACT

Exploratory Cycle Video

Exploratory Cycle Video

Target Audience

Students and alumni of Apple Developer Academies in Brazil, designers and SwiftUI developers interested in learning about Apple's native technologies and frameworks.
Senac students who want to rent a book from the library or book a sports court more quickly, avoiding queues and the time wasted downloading dedicated apps.

Exploratory Cycle Video

The Process

- Target audience research.
- Conversations with students and mentors.
- Research on engaging video formats.

The Decision

- Define the target audience.
- Adopt a WWDC-inspired format.
- Combine code demonstrations with real-life examples.

Artificial intelligence

Artificial intelligence

Exploratory Cycle - PoC

We created a personalized AI agent to act as a mentor, first identifying the learner's knowledge, preferences, and goals before developing a tailored learning path. Through step-by-step explanations, practical exercises, and adaptive feedback, the agent encouraged active participation rather than simply providing answers.
This experience showed us how AI can become a learning companion, supporting individual needs, independent thinking, and problem-solving.

Artificial intelligence

Exploratory Cycle - CoreNFC

We configured an AI tutor to support our learning of Core NFC through personalized explanations, practical examples, and real-world analogies. Its approach encouraged us to understand each concept through guided questions rather than simply receiving ready-made answers. We also established progress checks and short quizzes to reinforce knowledge, adapting explanations to our level and helping us learn at our own pace.
This experience showed us how carefully defined AI instructions can turn a general assistant into an interactive study tool, promoting deeper understanding, active reasoning, and knowledge retention.

Artificial intelligence

Exploratory Cycle - App Clip

For this cycle, we designed an AI tutor around a short, 20-minute learning session focused on App Clips. The agent began by exploring the learner's background, interests, and preferred learning methods, then created a personalized three-step study plan. Its approach combined conversational explanations, practical activities, and a final challenge designed to test whether the knowledge could be applied in a new situation.
This experience demonstrated how AI can transform a limited study session into a focused and interactive learning journey, balancing personalization, time management, and practical knowledge.



Obrigado

Challenge 6 - Grupo 5

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